

Pentatope Based Elliptic Curve Encryption for Privacy and Protection of Multimedia Data



D. Jayakumari, Ch. Rupa, and V. Nikhila

Abstract Now-a-days information security plays a vital role in this digitalized world. Data-breaching is the most serious threat in the current world due to the lack of proper security measures. In order to overcome such security issues, several cryptographic algorithms are developed. Most of these existing algorithms concentrated just on encrypting and decrypting process but not on the key size used and the computation time. To overwhelm this problem, a methodology was introduced by making use of the pentatope number in the elliptic curve cryptography, which tends to reduce the shared secrecy between sender and the receiver. It is used to provide security for multimedia data files such as text and image data present in various formats. In this ECC is used because it is the better alternative for the public key encryption compared to various encryption algorithms like RSA which offers a good level of security. The proposed methodology provides better security to the users in order to share their information in an ease manner.

Keywords Elliptic curve cryptography · Pentatope number · Information security · Multimedia files

1 Introduction

As the world became digitalized, the data can be transformed by a means of communication channel from anyone and from anywhere. The organizations located in different places in the world will share their multimedia data. They do their business by sharing their sensitive information through online [1]. Such information must be

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kept secure in order to avoid data breach issues. The multimedia data may present in different formats, either they may be text, image, audio or video etc.

We must keep these multimedia data files secure in order to avoid the unauthorized accessing of data. Several models are implemented in order to keep the data secure. Better method to keep our files secure is to use Elliptic Curve Cryptography for encryption compared to the other alternative encryption algorithms [2, 3].

Symmetric and asymmetric cryptographic methods are used to encrypt the data. In the symmetric key encryption method, single key is used for both encrypting and decrypting the data. It means same key is used by the sender to encrypt and receiver to decrypt the information. It is also named as secret key cryptography. The main issue to use this method is distribution of the secret key between the users [4].

In the asymmetric key encryption method, two different keys are used for encrypting and decrypting the data. It is also known as public key cryptography method, where public key and private keys are used for encryption and decryption of the data. It eliminates the problem of key distribution by providing a pair of keys in the process [5].

But it relatively takes more time to encrypt and decrypt data when compared to the symmetric key encryption method.

In the proposed methodology we make use of Elliptic Curve Cryptography (ECC) in order to provide better security to the multimedia files, because it uses the lesser key sizes to encrypt and decrypt the data [6]. It will help to reduce the computational time compared to the other cryptographic methods which makes use of large size keys. When the key size is large it becomes difficult to encrypt the large data files.

Multimedia security is provided by pentatope number-based public key infrastructure. We include this pentatope number in ECC in order to reduce the shared secrecy between the sender and receiver [7]. This is taken from the Pascal's triangle in order to reduce the usage of the global parameter values used in the conventional Elliptic Curve Cryptography method. It is the fifth number in any row of the Pascal's triangle based on the generator point (row) value [8].

2 Prerequisites

In order to implement our framework we are making using of some existing methods to provide security to the multimedia data such as text data and image data present in many formats like .txt, .doc, .pdf and .jpeg, .png, .tiff, .bmp etc.

2.1 Cryptography

Cryptography is the process of using some mathematical principles in order to encrypt and decrypt the data in a particular format. Thus, only the intended users who are transmitting the information can only understand and process the data. Here the

original text is converted into the cipher text by the sender and this is transmitted through a communication channel, so that no eavesdropper can understand what kind of information is passing over the channel and at the receiver end the cipher text is converted into an understandable format.

Symmetric Key Cryptography. In this method single secret key is used to encode and decode the data. It is also named as secret key encryption. The sender and receiver must know the secret key to cipher and decipher the data before transmitting through the communication channel. The main problem with this method is secret key distribution.

Asymmetric Key Cryptography. In this method a pair of keys are used in order to cipher and decipher the data. It is also known as public key encryption, where the public key is publicly available and the other key is kept secret. The data encoded using the public key can be decoded only using the corresponding secret key. It provides more security compared to the symmetric key encryption, but the computational time of this method is more compared to the other one. Encryption methods under this method are RSA, ECC, ElGamal, DSA etc [2, 4, 9].

2.2 *Elliptic Curve Cryptography*

Elliptic Curve cryptography is a public key encryption technique based on the elliptic curve theory [10–12]. It is a smooth curve in a predetermined field consisting of the points sustaining the Euler's equation. The keys used in this method are the product of finite points generated on the curve, these are of smaller size and are faster to encrypt and decrypt the data.

Conventional encryption methods consist of large keys because they are the product of the large prime numbers. As per the literature survey, ECC works 15 times better than the Ron Rivest Algorithm. A 256 bit key used in ECC provides more security than the 3072 bit key used in RSA.

2.3 *RSA (Rivest-Shamir-Adleman)*

RSA is a public key encryption method which makes use of the pair of keys in order to encrypt and decrypt the data. It is widely used to secure the sensitive data transmitted over an insecure channel. The keys are generated by the product of large prime numbers. The key size is more, so it takes more time to encrypt the large data files. ECC is preferred than RSA because point-based key generation is used in ECC whereas in the RSA modular operation is used in order to generate the keys [13].

3 Pentatope-Based ECC

In the proposed methodology, Elliptic Curve Cryptography is used including pentatope number. Which is used to reduce the number of external global parameters i.e., in the ECC model six global parameters are used in order to encrypt and decrypt the data, in our make the cryptographic algorithm makes use of just five global parameters. By using this cryptographic method, a generalized framework is implemented in order to encrypt and decrypt the multimedia data files present in different formats.

Figure 1 shows the pentatope-based ECC method which is used to protect the multimedia data files from the un-authorized users. It is helpful to the users who tends to transmit their data in a secure manner over insecure channel.

The curve parameters “ a ” and “ b ” values which must be less than the prime number considered, prime number “ P ”, private key of sender and receiver which can be of range 1 to $N-1$ (Where N is an integer value) these are the global parameters used in the pentatope-based ECC. As it comes under public key cryptography it requires both public and private keys for encrypting and decrypting the data. Private key of the contributor and receiver and the generator point are integrated in order to generate the public key of the end users transmitting the data. This public key is used in the process of encryption along with the pentatope number. Pentatope number is nothing but the fifth number in the row of a Pascal’s triangle which can be evaluated using the generator point value. It can be considered either from right to left or from left to right.

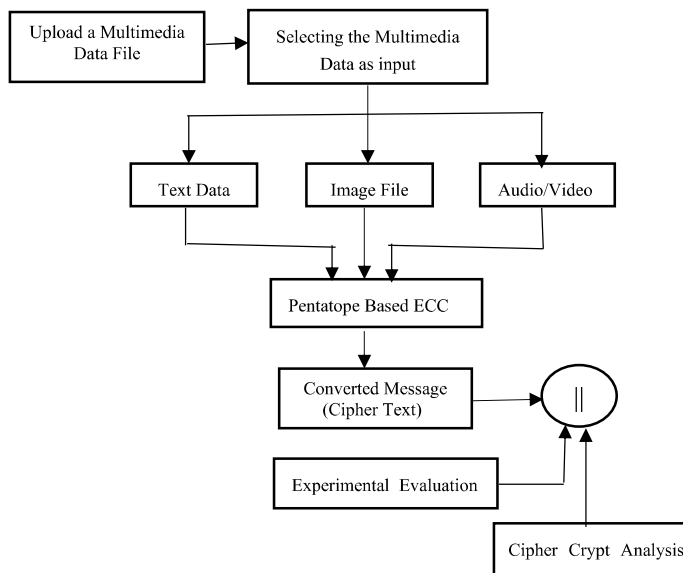


Fig. 1 Pentatope-based ECC

First an Elliptic curve is generated using the five external global parameters. From the generated curve we get some points. From these, one point is selected at random and this random point is considered as base point or generator point. Generator point consists of “x”, “y” values which can be denoted as $B(x, y)$. This generator point along with the private key of end users are used to generate the public key. Pentatope is considered in order to avoid the global external parameter (K) which is externally calculated in traditional ECC. By using this pentatope number, K value is internally calculated.

The obtained pentatope number is substituted in the Euler’s formula to obtain the value. This value is used in the encryption process. By using this, we can reduce the problem of shared secrecy between end users. This method provides more security to multimedia data files such as text and image files. The following are the main modules in the proposed methodology.

3.1 Pentatope-Based ECC: Key Generation Process

In this module we will evaluate the public key generation using the global parameters.

Step-1: Private Key ‘ P_{s1} ’ is selected randomly by the sender ranges from 1 to $N-1$, where N is the natural number.

Step-2: Sender’s Public Key ‘ P_{p1} ’ is generated by using the private key of sender and generator point ‘ B ’.

$$P_{p1} = P_{s1} * B$$

where B : Generator point $B(x, y)$

P_{p1} : Public key of sender

P_{s1} : Private key of sender.

Step-3: Private Key ‘ P_{r1} ’ is selected randomly by the receiver ranges from 1 to $N-1$.

Step-4: Receiver’s Public key ‘ P_{p2} ’ is generated by using the private key of receiver and generator point ‘ B ’.

$$P_{p2} = P_{r1} * B$$

where B : Generator point $B(x, y)$

P_{r1} : Public key of receiver

P_{p2} : Private Key of receiver.

3.2 Pentatope-Based ECC: Encryption Process

In this module the encryption process is explained, here it will take the input from the sender and encrypts the plain text received from the sender into cipher format which can be transmitted over insecure channel. Here the inputs are Generator point “ B ”, public key “ P_{p2} ” and the plain text “ m ” and output is a Cipher text.

Here we are using the pentatope number as a key to reduce the shared secret. Formulae to calculate nth pentatope number.

$$(S*(S + 1)*(S + 2)*(S + 3))/24.$$

Step-1: Let the message encrypted is to be 'm' and it has a point 'M' on the elliptic curve.

Step-2: Pentatope number is taken from the level equal to the 'x' value of generator point from Pascal's triangle.

$$S = \text{Pentatope number}(x).$$

Step-3: Here we are using Euler's function to calculate the Key 'K' by using pentatope number.

$$K = s^2 + s + 41$$

where s: pentatope number.

Step-4: Then two cipher text points will be generated i.e. C_1, C_2 .

$$C_1 = K * G$$

$$C_2 = P_m + K * P_{p2}$$

where K: key generated using pentatope number

G: Generator point

P_m : Message Point

P_{p2} : Public key of receiver.

3.3 Pentatope-Based ECC: Decryption Process

In this module the encrypted data is decrypted using the private key, here it will convert the transmitted cipher text to its original format.

Here the input is taken as the Cipher text and the output is the original plain text sent by the contributor. The process undergoes by the following steps:

Step-1: This decryption is obtained by using the Cipher text C_1 , Cipher text C_2 , Private key of receiver.

Step-2: The product of C_1 and Private Key of receiver is computed by the receiver.

Step-3: Then the product is subtracted from point C_2 by the receiver and we get it as.

$$M = C_2 - (P_{r1} * C_1).$$

where P_{r1} : Private key of receiver.

4 Results and Crypt Analysis

In this section, how often our generalized framework will encrypt and decrypt the multimedia data is demonstrated. In order to test our proposed model various formats of multimedia data are considered as input. It is tested on both the text data formats and image data formats.

Some of the text data formats considered in our model are .txt, .doc, .pdf and image data formats are .jpeg, .png, .bmp, .tiff. The test analysis of proposed model is shown in this section.

Table 1 demonstrates the performance of the Elliptic Curve Cryptography and the Pentatope based ECC. It is tested on the text data present in .txt format and it depicts that the ECC takes relatively more time to encrypt and decrypt the text data when compared to the proposed pentatope-based ECC.

Table 2 demonstrates the performance of the ECC and Pentatope-based ECC on the image data. It is tested with the various sizes of image files. It shows that the pentatope-based ECC takes relatively less time to encrypt and decrypt the data when compared to the traditional ECC.

Table 3 demonstrates the performance of the ECC and Pentatope-based ECC on the image data file formats (.png, .jpg, .bmp, .tiff). It is tested with the various sizes of image files. It shows that the pentatope-based ECC takes relatively less time to encrypt and decrypt the data when compared to the traditional ECC.

Figure 2 shows the performance analysis plot for the text data files of different sizes. It represents the time taken to encrypt and decrypt the text files of different sizes. It depicts that the ECC takes comparatively more time when compared to ECC.

Figure 3 shows the analysis plot for the ECC and pentatope-based ECC in case of .png files. It indicates that the ECC takes more time to encrypt and decrypt the files when compared to the proposed model.

Figure 4 shows the performance of the pentatope-based ECC in order to cipher and decipher the .bmp image files which makes the files more secure while transmitting them through an insecure channel.

Table 1 Text data performance

File size (kb)	ECC (time in sec)		Pentatope-based ECC (time in sec)	
	Encryption	Decryption	Encryption	Decryption
10	0.6213	0.05856	0.00117	0.00085
20	0.4842	0.05635	0.00057	0.01056
30	0.7650	0.57355	0.00062	0.00102
40	0.8243	0.05822	0.00072	0.00256
50	1.83	0.05940	0.00095	0.01010

Table 2 Image data performance

File size (kb)	ECC (time in sec)		Pentatope-based ECC (time in sec)	
	Encryption	Decryption	Encryption	Decryption
17.4	10.0	10.32	3.7234	2.4123
28.8	10.1	10.56	5.1080	3.7129
59.6	10.7	10.56	7.3234	4.1673

Table 3 Different image data formats performance

File size (kb)	Image files	ECC (time in sec)		Pentatope-based ECC (time in sec)	
		Encryption	Decryption	Encryption	Decryption
17.8	.png	5.0905	5.1163	0.00396	0.00660
	.jpg	6.5837	6.6762	0.00543	0.00909
	.bmp	7.1999	7.0930	0.00960	0.00080
	.tiff	7.8635	6.8856	0.00056	0.00177
86.8	.png	6.9564	6.7067	0.01320	0.01452
	.jpg	7.1276	7.2652	0.01357	0.01276
	.bmp	7.9546	7.1423	0.00573	0.00436
	.tiff	6.5243	6.1452	0.01525	0.01136
170.2	.png	7.9542	7.5623	0.00173	0.00291
	.jpg	8.2314	8.1632	0.00413	0.00231
	.bmp	7.2652	7.1732	0.00653	0.00025
	.tiff	7.1423	6.2634	0.00976	0.00653

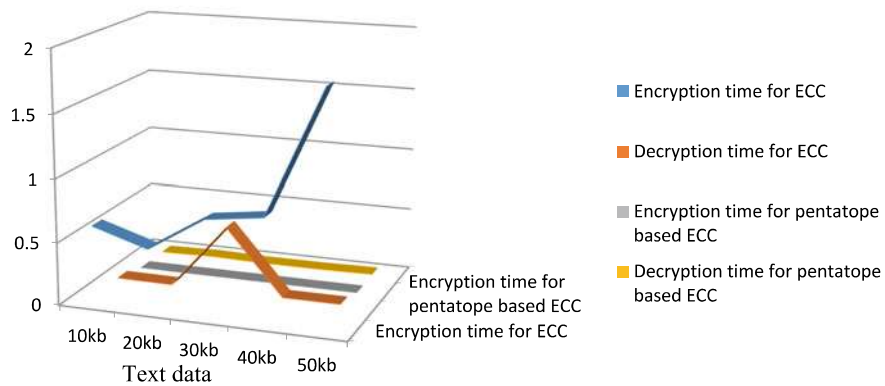
**Fig. 2** Performance analysis of the text data

Figure 5 represents the plot for the ECC and pentatope-based ECC while encrypting and decrypting .tiff image files. It demonstrates that ECC takes more time when compared to pentatope-based ECC.

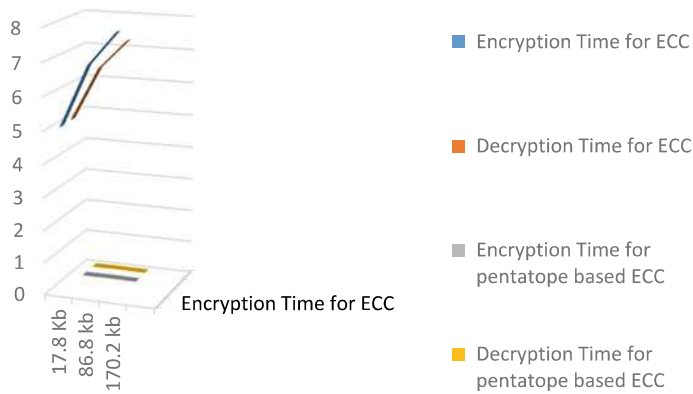


Fig. 3 Performance analysis of the .png image

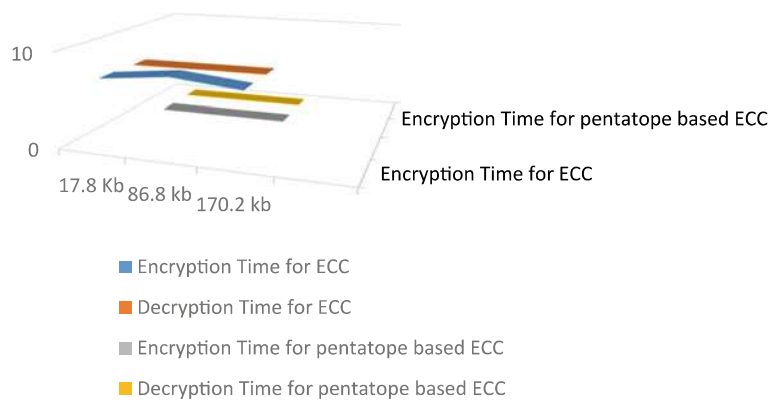


Fig. 4 Performance analysis of the .bmp image

5 Conclusion

The proposed comprehensive cipher, Pentatop-based ECC is used to reduce the cost factor in terms of external global parameters that are given in conventional ECC. This cipher is tested on many formats of the text data (.txt, .doc, .pdf) and image data (.png, .bmp, .tiff, .jpg) and it is proved that pentatope-based ECC reduces the time complexity and space complexity which leads to increase the efficiency of the cipher. In future we would like to test this cipher on other multimedia data such as audio and video files.

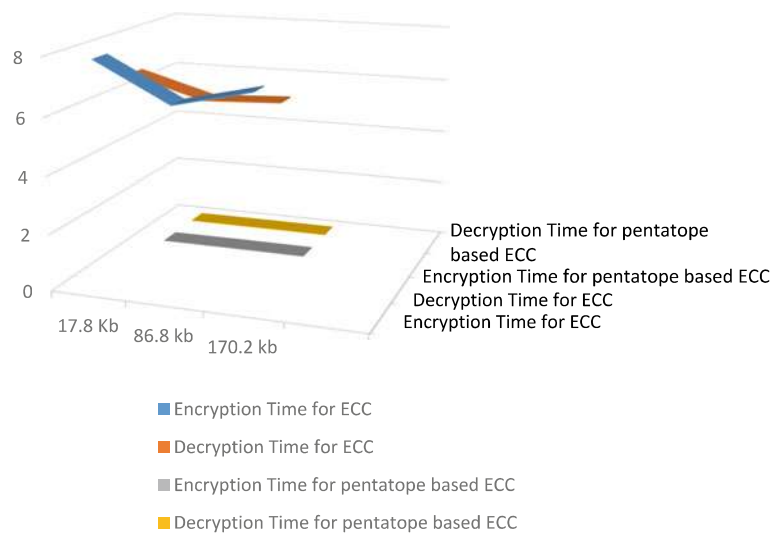


Fig. 5 Performance analysis of the.tiff image

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